HOMOLOGY FROM A FINITE SEMI-SEGAL PRESENTATION

The experimental root `unit/sag/hcat-segal/All.seg` constructs new composition and associativity operations from horn fillers, proves their pentagon, and checks the existing homology theorem files against this interface. The original `hcat` development is unchanged. This establishes the proposed refactor at the level of an internal category supplied with finite semi-Segal data. Constructing these data from the original ambient categorical axioms remains a separate task.

1. The inputs and the selected operations

For an object type O and an edge family $E : O \to O \to \mathcal{U}_l$, take a triangle family

$$A_2(f,g,h) : \mathcal{U}_l, \qquad \mathsf{isContr}\left(\sum_{h:E(a,c)} A_2(f,g,h)\right) \quad (f : E(a,b),\ g : E(b,c)).$$

Write fg for composition in the order of traversal. Define fg and its triangle to be the two components of the selected horn filler. The fundamental theorem of identity types gives, for every h,

$$A_2(f,g,h) \simeq (fg = h).$$

This equivalence justifies using composition paths as the triangle family at the next stage. The normalized horn fillers are singletons.

A Λ^3_1 horn now consists of four vertices, six edges, and three paths q_{012}, q_{123}, q_{013}. Its missing triangle is

$$q_{023} : e_{02}e_{23} = e_{03}.$$

Take a family $A_3(u, q_{023})$ over these full boundaries, with contractible paired horn fillers. On the canonical horn of f, g, h, put

$$e_{02} = fg, \quad e_{13} = gh, \quad e_{03} = f(gh), \qquad q_{012} = q_{123} = q_{013} = \mathsf{refl}.$$

The missing triangle selected by the horn condition is

$$\alpha(f,g,h) : (fg)h = f(gh).$$

This is `Associativity.hcomp-assoc`. No associator is an input parameter, and no comparison with the former associator is used.

2. Normalized tetrahedra

For a complete triangular boundary define

$$L = \mathsf{ap}_{(-e_{23})}(q_{012}) \cdot q_{023},$$
$$R = \big(\alpha(e_{01}, e_{12}, e_{23}) \cdot \mathsf{ap}_{(e_{01}-)}(q_{123})\big) \cdot q_{013}.$$

Both paths go from $(e_{01}e_{12})e_{23}$ to e_{03}. The normalized tetrahedron family is $(L = R)$.

For a fixed horn, its paired fillers are the fibre of concatenation by $\mathsf{ap}_{(-e_{23})}(q_{012})$. Concatenation by a path is an equivalence, so this fibre is contractible. Thus normalization retains the required horn condition without any truncation assumption on E.

Moreover, it is equivalent to the original A_3 family over each boundary. To see this, eliminate the three given triangle paths. On the canonical horn, the missing triangle selected by A_3 is exactly the defining associator, and the normalized cell is its left unit path. This gives a normalized

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cell at the original selected missing face for every horn. Both families have contractible total spaces over the same missing-face type and the same chosen base point. Applying the fundamental theorem of identity types to both gives their fibrewise equivalence. These steps are checked in `SemiSegalNormalize`.

3. The pentagon is a missing tetrahedron

The final input is a family A_4 over complete boundaries made from these normalized tetrahedra. Require contractibility of

$$\sum_{t_{0234}} A_4(u, t_{0234})$$

for each Λ^4_1 horn u. This is a condition on all matching boundaries, not a separately specified associahedral equation.

For four composable edges f, g, h, k, form the horn with edges

$$e_{02} = fg, \quad e_{13} = gh, \quad e_{24} = hk, \quad e_{03} = f(gh), \quad e_{14} = g(hk), \quad e_{04} = f(g(hk)).$$

Six triangles are reflexivity paths. The remaining four are

$$\begin{aligned} q_{023} &= \alpha(f, g, h), & q_{134} &= \alpha(g, h, k), \\ q_{024} &= \alpha(f, g, hk), & q_{034} &= \alpha(f, gh, k) \cdot \mathsf{ap}_{(f-)}\alpha(g, h, k). \end{aligned}$$

The four supplied tetrahedra are left unit paths. The missing tetrahedron, selected by the 4-horn condition, says

$$\mathsf{ap}_{(-k)}\alpha(f, g, h) \cdot \big(\alpha(f, gh, k) \cdot \mathsf{ap}_{(f-)}\alpha(g, h, k)\big) = \alpha(fg, h, k) \cdot \alpha(f, g, hk).$$

Reverse this equality and reassociate path concatenation to obtain the exact `HcompPentagon.Pentagon` interface. This is the checked theorem `SemiSegalPentagon.pentagon`.

This finite construction implements the low-dimensional semi-Segal method, with normalization made explicit; compare Capriotti–Kraus, *Univalent Higher Categories via Complete Semi-Segal Types*, Section 3, `https://arxiv.org/abs/1707.03693`. The simplicial boundary organization is compatible with the alternative ambient presentation discussed by Sattler–Wärn, but no ambient degree-5 construction has been implemented here.

4. Units and the homology interface

Identity edges and two degeneracy triangles are additional inputs: $A_2(1, f, f)$ and $A_2(f, 1, f)$. They give the two unit laws by the triangle equivalence. This is the unit fragment used by the homology proofs. We do not assert a full degeneracy structure, completeness, or Rezk univalence for this experimental presentation.

The names `Composition`, `CompLaws`, `Associativity`, and `HcompPentagon` expose the interface expected by the existing theorems. A small inverse interface supplies the generic implication from two-sided inverse to invertibility of postcomposition. In `Homology.WithPentagon`, `hpent` is a definition using the new pentagon theorem, not a segment parameter.

The existing theorem sources are reused by relative links. They cover copowers, tensor maps, homotopy invariance, dimension, additivity, Fubini, excision, groups, the positive-degree Mayer–Vietoris sequence, reduced suspension, and the existing sphere calculations. The copower and stability assumptions retain their universal-property formulations. In particular, for a pushout of types, excision and stability supply the pullback from which the sequence is constructed.

All three repeating positive-degree exactness positions use the same structural boundary and group structures within this interpretation. Degree-zero compatibility and categorical naturality remain open, and no exactness is asserted at the final H_1 term. Sphere comparisons retain their existing scope as bijections of sets.

5. What has and has not been replaced

The original 169 `hcat` source files are unchanged. In that root, `hpent` is still assumed. In the experimental root, its replacement is derived for the new operations and the homology proofs recheck. No equality between the old and new group structures or connecting maps is claimed: that would require comparison data relating the two categorical presentations.

The experiment has explicit finite semi-Segal parameters, alongside the unchanged foundational parameters `univ`, `images`, `surjections` and the homology universal-property inputs. No new axioms have been installed in the original development. The index type called `Cat` in the experiment merely indexes internal presentations; it carries no ambient functor composition. There are no judgmental associativity or unit rules for categories, no UIP assumption, and no second equality.

The next task is to construct the finite internal data from an ambient coherent category presentation, or to supply a model directly. The refactor has removed the need to preserve the old associator in order to rebuild homology. It has not removed the need for categorical coherence itself.